# Where Does Evolvability Reside? Organisms, Encounters, and Ecological History in Self-Reproducing Cellular Automata

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## Abstract

Evolvability concerns the capacity to generate useful heritable variation. In spatial artificial life, that capacity can depend on an organism's inherited state, the organisms it encounters, and ecological histories that change both. We frame these as compatible causal contributions and implement a reproducible encounter intervention in the original nine-state evoloop cellular automaton. An exhaustive comparison of all 59,049 local neighborhoods establishes agreement with two reference routes. A documented seed reproduces, and deterministic replay and checkpoint/resume agree. Natural exploratory variation supplies a second verified input organism through a transient-to-stable inheritance path. We prospectively freeze 32 main crossed encounters, eight held-out encounters, 40 separated controls, and 24 alone controls, with complete starting grids and conservative pair-level ancestry. All 104 atlas states completed. Candidate signal novelty occurred in 6/40 near configurations and 0/64 separated/alone controls. Eleven isolation paths supplied no confirmed stable novel transmission under the frozen criterion. A candidate interaction contrast of +0.375 in the main panel changed to −0.500 in held-out configurations; neither panel showed an inherited-novelty interaction. A paired ecological benefit assay is frozen but remains pending viability calibration. The organism-transfer and ecological-history hypotheses remain untested.

## 1. Introduction

Self-reproducing cellular automata make evolutionary events inspectable at the level of individual cells. Rules can remain fixed while interacting structures produce inherited changes. Sayama's evoloop demonstrates this possibility in a deterministic, synchronous, nine-state lattice: reproductive arms interact, offspring differ, and competition for space changes population composition [@sayama1999evoloop]. The relevant evolutionary unit is nevertheless not obvious. A circulating signal sequence controls a loop's construction behavior, but the sequence it passes to a descendant can depend on structures encountered during reproduction.

Our question is where the capacity for further adaptive change resides. Three compatible hypotheses organize the problem. H1 concerns transferable organism properties: organisms sampled from evolved populations should retain higher beneficial-heritable-variant yield than matched ancestors under the same held-out partners, geometries, and phases. H2 concerns encounters: crossing focal and partner identities should reveal organism-by-partner interactions, possibly including partner-dependent rankings. H3 concerns ecological history: history can change inherited organisms and the distribution of encounters they subsequently experience. The hypotheses need not compete for an exclusive explanation; an organism can carry a variation bias that is expressed differently against different partners, with both shaped by its prior ecology.

History is upstream of the present state. Under deterministic CA dynamics, identical complete starting states and intervention schedules imply identical futures. A history transplant can identify paths mediated by organism state or encounter distribution; it cannot establish an additional historical force after holding the entire state fixed. This constraint is useful because it specifies which comparisons can distinguish levels of causation.

The project has a limited potential connection to recursive self-improvement: an evolving system might acquire greater capacity to generate further adaptive change. Neither ordinary adaptation nor greater population size demonstrates that claim. We first measure the production and inheritance of variation under a declared encounter panel. Ecological benefit requires a separate comparison; increased evolvability would then require a comparison of such beneficial variation yields between transferred organism states or histories.

This checkpoint therefore targets the smallest reliable intervention addressing H2. It establishes verified dynamics, freezes a small crossed atlas, and follows candidate novelty through explicit isolation assays. It retains lethal and nonproductive encounters and marks unresolved cases. It does not infer organism-level mutation rates from ambiguous pedigrees, or generalize from one selected descendant to independently evolved populations.

## 2. Related work and contribution

The original evoloop study introduces structural dissolution, interaction- mediated genotype change, and spatial selection within a deterministic rule system [@sayama1999evoloop]. Its configurations include arm takeover, smaller descendants, and transient or rectangular structures. The recent review situates these models among self-reproducing CA and distinguishes reproduction supporting inherited variation from exact replication [@sayama2025review; @sayama2024reviewpreprint]. Our study uses the original synchronous rules; it does not combine self-protecting, asynchronous, pathogen, or sexually reproducing variants with that baseline.

Subsequent work established a much richer genotype space than size alone captures. Salzberg and Sayama report genotype-dependent colony morphology, mutation bias, persistence, and population density, including a round-robin competition among all 15 size-4 genotypes [@salzberg2004complex]. They also describe inherited mutational biases. Thus neither crossed competitors nor inheritance of a variation bias is by itself our proposed novelty. Their population competition endpoint addresses which types persist; our immediate endpoint is whether specified encounters produce new genotypes that reproduce after separation from the encounter.

Salzberg, Antony, and Sayama analyze genealogy, biased accessibility of larger species, pathogens, and periodic localized or distributed block removal [@salzberg2004hostile]. Their treatment of transient species is particularly important: structures that do not directly reproduce their own genotype may produce stable descendants. Discarding them at first observation would exclude an established route to inherited variation. The present isolation search therefore allows a candidate to produce a different stable descendant before checking repeated transmission. Ecological disturbance and limited complexity growth are prior results, not outcomes to be rediscovered here under a new name.

The intended addition is a controlled, prospectively specified comparison of variation-generating outcomes: candidate production, isolated inheritance, and ultimately paired ecological advantage over both encounter inputs. Exact starting states, negative encounters, horizon censoring, and ambiguous ancestry make the denominator reviewable.

Held-out configurations test whether an interaction contrast transfers across geometry and phase, while future organism and encounter-distribution transplants address its causal location. At this checkpoint, novelty is an intervention-and-measurement comparison rather than a claim of a new physical mechanism.

# 3. Methods

## 3.1 Rule system, reference agreement, and licensing

Each cell has one of nine states. The local neighborhood is ordered center, top, right, bottom, left (CTRBL); updates are synchronous and boundaries are periodic. Explicit rules are completed under rotations of the four neighbors, then dissolution and undefined-neighborhood defaults supply a full lookup table. We select the original evoloop transition system implemented by CellPyLib at commit 743e936d48f8520f6f4ac652570ac7bb46414189 [@antunesCellPyLib]. The expanded uint8 lookup-table SHA-256 is 9d0652aaceb55dd3c7ff9814b45ae56c7782c8a20cab0fc88e11fc6d1cae37f3.

An independent expansion of the author's EvoLoopS 1.0 rule file agrees on all 59,049 neighborhoods. The documented CellPyLib species13 seed matches the author's 13.img; it differs from the author's default 13-2.img seed. Seed variants do not imply different transition rules. We do not claim to reproduce the original default evolutionary demonstration. The reference modules are redistributed under Apache License 2.0 with their license and attribution; the author software, seed catalogue, artwork, and PDFs are not redistributed. Source hashes, publication-year caveats, and reuse decisions appear in docs/source_review.md [@cellpylibEvoloopDocumentation; @sayamaEvoLoopS].

## 3.2 Engine and focused verification

A headless C kernel accessed from Python uses compact uint8 arrays and a 59,049-entry lookup table. Two arrays implement synchronous updates. No mutation injection occurs in the atlas. A separate occupied-cell provenance layer records potential input contribution without modifying the CA state. Complete initial grids, rule/configuration hashes, event times, snapshots, and final hashes are saved. Normal runs stream structural metrics and retain selective candidate evidence rather than exhaustive frame histories.

Verification exhaustively compares local transitions with the callable reference, checks a random periodic grid against a cellwise oracle, observes documented seed reproduction, and compares exact deterministic replay and checkpoint/resume. Rotation tests cover square and naturally generated rectangular genotypes. Measurement development found that body-only crops can destroy active arms; we rejected that extraction method before freezing the main atlas. The final assay preserves complete detached components. The runtime benchmark and peak resident memory are recorded in data/verification.json; one worker is used because available system memory, rather than CPU count, limits concurrency. During acquisition, a read-only compiled contact diagnostic passed 6,760 equivalence cases, including four complete raw trajectories, before accelerating the remaining controls. The frozen physics and outcome definitions were unchanged; execution.json records the switch. Near isolation assays ran separately once near acquisition was complete, within measured memory headroom.

## 3.3 Exploratory panel and prospective encounter configurations

The panel has two organisms: S13, the documented seed, and N01, the first new stable genotype confirmed in a 160×160 exploratory torus (EXP-discovery-001). Exploration starts with one S13 seed at (65,45), samples every ten updates, and stops after two verified genotypes. Up to four naturally observed detached phases per candidate are available for discovery testing. N01 is obtained through a transient-to-stable path; the catalogue

stores its confirmed descendant state. This is one selected exploratory history, not independent evolutionary replication.

Protocol 001 and configs/atlas-v1.json are frozen before the main batch. Each input snapshot is advanced alone by 0, 30, or 60 updates, retaining its entire occupied configuration. The resulting phase states must each contain one recognized core with unchanged genotype and no boundary contact. The offsets are elapsed updates relative to different snapshot origins, not homologous reproduction phases. Complete grids make this distinction reproducible.

All four ordered pairs are crossed against four main relative geometries at phases 0 and 60 (eight states per pair), and two held-out geometries at phase 30 (two per pair). The main torus is 384×384. Rotations are exact multiples of 90 degrees, bounding rectangles do not overlap, and the protocol records coordinates. Forty matched separated controls retain identities, phases, orientations, and row offsets in a 640×640 torus with horizontal displacement 320 cells. Their lack of contact is verified throughout the window. Twenty-four alone controls cover each organism, phase, and orientation. Every state is observed for 1,600 completed updates.

## 3.4 Contact, structures, genotypes, and ancestry

Input occupied cells receive provenance masks 1 and 2. A newly occupied cell inherits the union of occupied contributors in its CTRBL neighborhood. This is a causal overapproximation, not a genetic pedigree. Operational contact is the first update with occupied exclusive-provenance von Neumann adjacency or mixed occupied provenance. Contact is checked every update and records grid time. It does not by itself prove a mutational mechanism.

The detector samples enclosed rectangular zero cores with intact inner sheaths in states 2/3 every ten updates. This is a structural observation, not a viability claim. For channels containing only 0/1/4/7, the operational genotype is the clockwise signal-ring word, including carrier spacing, normalized by cyclic shift without reflection and paired with sorted core dimensions. It preserves chirality and distinguishes gene spacing. It is not the published counterclockwise, birth-anchored species identifier [@salzberg2004hostile]. Contaminated channels have unknown genotype.

Structures are tracked by core coordinates. First appearance and loss times are sampled proxies, interval-censored within ten updates. Input-core persistence is reported as structural survival; it does not establish continued viability or lineage survival. Boundary touch is flagged because the rectangular detector does not join structures crossing the torus seam. Nonrectangular and inseparable structures remain possible sources of missed candidates.

Novelty requires a genotype different from both encounter inputs. Snapshots retain the source grid, extraction box, and provenance. After contact, genetic parentage is conservatively ambiguous, so the primary unit is pair-level novelty. A changed structure, a changed signal genotype, a detached organism, a heritable variant, and a beneficial variant are separate outcomes.

A reporting audit added during acquisition, before inheritance analysis, clarifies that the code field genotype is a composite signal-word/core-shape identity. The frozen primary endpoint retains this definition and is reported as composite organism-identity novelty. A separately labeled sensitivity endpoint requires the channel word alone to differ from both inputs. Shape-only differences are morphological candidates, not genetic changes. No raw outcomes or search budgets change; protocol 002 records the timing and reason for this

audit. The signal-word result must not be described as prospectively frozen.

## 3.5 Isolation and transient-to-stable inheritance

Isolation is an explicitly imposed intervention. We extract the full occupied connected component, including arms, only when it contains exactly one recognized core and avoids the grid edge. We remove other components and center the snapshot in an empty 192×192 torus. Oversized, shared, or boundary-spanning components are ineligible. The earliest two distinct extractable snapshots per run/genotype constitute the frozen search budget; size eligibility is checked after capture, so oversized attempts consume that budget and remain unresolved. Later phases and siblings may be missed.

Each isolation generation runs for up to 1,600 updates. We seek the first newly positioned closed core with a valid genotype and a complete detached component. The first descendant may differ from the encounter candidate. It is then isolated and must reproduce its own genotype. If the initial candidate was transient, a further generation must transmit the stable genotype, providing two stable transmissions. Closure-first-observed and detachment-first-observed times are separate. Stable genotypes arising only after isolation are explicitly assay-derived, and novelty is checked again against both encounter inputs.

A path that fails to reproduce by the horizon is unresolved. An empty isolation grid is assay extinction, not proof of ecological nonviability of the original structure. The first-descendant search is conservative: later offspring might reproduce successfully. Stable isolated inheritance establishes a reproducible transmission path through the stated intervention, not ecological independence of every structure with the same observed genotype.

## 3.6 Benefit assay and finite-panel analysis

Before selecting any atlas winner, we freeze a paired disturbance assay in 192×192 tori over 1,600 updates. At updates 600 and 1,000, a specified 24×24 block is removed; this is an external ecological intervention. The metric is mean intact-core population of isolation-confirmed genotypes, sampled every 20 updates. Two development and two independent confirmation schedules specify block positions and initial orientations. A candidate must exceed both input means in both panels. The measure targets persistence and recolonization occupancy, not evolvability by itself. Because recognized cores can include doomed individuals, viability calibration is required before confirmed-benefit claims. Benefit remains pending if those assays or calibration are unfinished.

The primary completed endpoint is any confirmed novel stable inheritance path per near configuration, with all 32 main and eight held-out states in the denominator. Candidate-only and unresolved outcomes remain explicit. A 2×2 focal-by-partner rate matrix and difference of differences summarize interaction: (S13,S13 − N01,S13) − (S13,N01 − N01,N01). Main and held-out panels are analyzed separately, along with signs of the partner-specific focal contrasts. These are pair-configuration interactions under ambiguous ancestry, not identified focal mutation capacities. A partner main effect is insufficient for H2.

Distinct complete deterministic configurations are finite-panel census units. Repeated states are reproducibility checks. We report no binomial confidence intervals, p-values, or resampling of repeated deterministic trajectories. Held-out geometries test configuration transfer, not transfer across independent organisms or ecological histories. Conditional beneficial-heritable yield is unknown until benefit confirmation is completed; it must not be plotted as zero.

# 4. Results

## 4.1 Verified foundations and exploratory inheritance

All 59,049 local transitions agree with the selected callable reference and the independent author-rule expansion. The seed agrees byte for byte with the documented author seed, reproduces into multiple closed cores, and passes exact replay and resume checks. The exploratory candidate observed at update 2,910 has an 11×13 inner core and a changed signal word. In complete-component isolation it produces a stable 12×12-core descendant; two subsequent isolated generations reproduce that stable genotype. Detachment is first observed at 380, 390, and 390 updates in the three assays, sampled every ten updates. N01 denotes the stable descendant, not the transient source configuration. The transient and stable structures share the changed signal word: their 11×13-to-12×12 difference is morphological stabilization, while their shared signal word differs from S13's. This separates phenotype change from signal genotype change in the exploratory path.

## 4.2 Encounter atlas

All 40 near configurations completed the 1,600-update window with recorded contact and no boundary-touch flags. Candidate composite identity changes occurred in five of 32 main states and one of eight held-out states. All seven run/identity candidates also differed in signal word from both inputs, so the strict reporting audit currently agrees with candidate classification. Five candidate-producing configurations supplied 11 complete-component snapshots; one candidate could not be separated within the window. None of the 11 isolation paths met the frozen stable-transmission criterion. Six produced a changed detached descendant in the first isolation stage. Four of those next isolated descendants dissolved; two did not reproduce the same identity by the next horizon. Two initial snapshots dissolved, and three remained without a qualifying detached descendant. These are completed assay observations, not proof that all possible ecological inheritance paths are absent.

The 40 matched separated states had no contact and no candidate novelty. All 24 alone controls likewise had no candidate novelty. No atlas run touched a boundary during observation. The matched near-minus-separated candidate rate was 6/40 = 0.150; this is a descriptive finite-panel contrast. Full initial/final hashes and candidate extraction checks pass for all 104 states. Four chunked encounter replays reproduce final grids and provenance, and 17 isolated-generation replays match recorded terminal hashes.

| Focal × partner | Main candidate / n | Held-out candidate / n | Stable inheritance |
|---|---|---|---|
| S13 × S13 | 3/8 | 0/2 | 0/8; 0/2 |
| S13 × N01 | 0/8 | 0/2 | 0/8; 0/2 |
| N01 × S13 | 1/8 | 1/2 | 0/8; 0/2 |
| N01 × N01 | 1/8 | 0/2 | 0/8; 0/2 |

In the main panel, the candidate-only focal contrast was +0.250 with S13 as partner and −0.125 with N01 as partner, a rank reversal and difference of differences +0.375. In held-out configurations the corresponding contrasts were −0.500 and 0.000, giving −0.500 and no strict rank reversal. The candidate pattern therefore did not transfer. Confirmed isolated-inheritance contrasts were zero in both panels because no path satisfied that criterion. Candidate interactions do not establish H2 for useful heritable variation.

Thirty of 40 near states lost both recognized initial cores by the horizon, but this is not equivalent to lineage extinction: many retained numerous input-type descendants. Five

near states had no recognized closed core at the horizon; one was completely empty, while four retained debris. Original cores also disappeared in all separated and alone controls while colonies continued. These observations reinforce the distinction between individual structure persistence, reproduction proxies, and lineage viability.

Summed recorded run time was 1,177.1 seconds (19.6 minutes), with peak reported runner RSS 63,400 KiB (61.9 MiB), excluding separate rendering processes. Raw results, unresolved outcomes, and controls are in data/atlas/outcomes.json and analysis.json. Figures show the observed-versus-confirmed funnel, crossed matrices, and three raw-grid replays. Benefit assays and viability calibration remain pending; beneficial-heritable yield is unknown, not zero. H1 and H3 remain untested, and no increased-evolvability or RSI claim is supported.

### 4.3 Exploratory extension of a multi-transient path

A follow-up selected after observing the failed first-descendant inheritance search is reported separately as EXP-lineage-followup-001 (protocol 003). Starting from the earliest N01×N01 offset candidate at encounter t=1,550, it allows up to six serial isolation stages and requires two consecutive stable transmissions. The 3×12-core source produces a 6×6-core descendant, which produces a 4×4-core descendant; that descendant dissolves in the third stage. No stable path is confirmed. This exploratory diagnostic leaves the frozen atlas outcomes unchanged and provides no independent confirmation sample.

## 5. Limitations and alternative explanations

The panel contains one documented organism and one selected descendant from one exploratory history. Selection for isolation eligibility can favor certain structures or phases; genotype is less than the complete transferred organism state. Phase, arm state, phenotype, and genotype effects are entangled in this first atlas. A measured pair interaction need not be a gene-sequence interaction. Equal global observation horizons also give unequal postcontact exposure when contact latency differs. The operational genotype may miss junction-state distinctions or identify states with different future behavior as the same circulating signal word.

Contact provenance overapproximates causality. Structure detection does not provide universal individuality, precise biological births, or genetic pedigree. Isolation removes ecological assistance and can stabilize or disrupt a structure. Its limited first-descendant and phase search can miss viable paths. Boundary and nonrectangular structures can be missed. Surviving closed cores are therefore distinct from independently verified viable individuals.

Separated controls change grid extent as well as displacement, so they support contact-window diagnostics rather than an unrestricted causal estimate of contact alone. Relative geometry, orientation, and phase are interventions, not random samples from a natural encounter distribution. Descriptive contrasts cannot establish population-general effects. Beneficial variation, increased evolvability, H1, H3, and recursive self-improvement remain unestablished.

## 6. Subsequent experiment plan

The immediate next experiment is paired candidate-fate calibration, before expanding the atlas or interpreting failed isolation as intrinsic nonviability. Replay the six candidate-producing states, compare native continuation with isolation at fixed maturation delays, and follow a predeclared bounded set of transient offspring paths. Include S13/N01 positive controls under the same extraction schedule. This holds the recorded source

trajectory fixed while changing surroundings and extraction time, testing ecological dependence and assay sensitivity. It cannot recover unambiguous native parentage merely from pair-level provenance. Budgets and criteria must be frozen before execution; see docs/next-experiment.md.

For H1, evolve independent population histories, retain matched ancestors, and sample transferable complete organisms using a fixed sampling rule. Cross each with the same held-out partner/geometry/phase panel. This transplant holds the assessment encounter distribution fixed and changes inherited organism state. Population history, not each encounter or repeated deterministic run, is the replication unit. Compare conditional beneficial-heritable yields only after calibrated inheritance and benefit assays.

For H3, compare undisturbed histories, structured disturbances, and shuffled controls matched on disturbance amount and an explicitly declared shuffle unit. One transplant puts history-derived organisms into a shared encounter panel, holding encounter distribution fixed. A second puts shared organisms into history-specific encounter distributions, holding organism state fixed. Crossing the two separates inherited-state and encounter-distribution pathways and their interaction. Preserve independent histories as replication units. All histories still act through present state and future scheduled interventions; no residual history effect is expected for identical complete states.

## Reproducibility and data availability

Repository: https://github.com/ReloadLightly/evoloop-levels. Reproduction commands: python -m scripts.verify, python -m scripts.atlas run, python -m scripts.atlas inherit, python -m scripts.analyze, python -m scripts.figures, python -m scripts.render_manuscript, and python -m scripts.check_evidence. README explains fresh versus resumed runs. Protocols, configurations, source, compact raw grids, analysis, and figures are versioned. Each figure must reference experiment IDs and a regeneration command. The manuscript and bibliography are working artifacts, not a submitted paper.

## References

Verified BibTeX is in paper/references.bib. Primary papers are linked in docs/source_review.md; bibliographic caveats and exact inspected hashes are recorded there. The review's final journal issue is 2025, while its preprint is 2024. No third-party PDFs are included in this repository.

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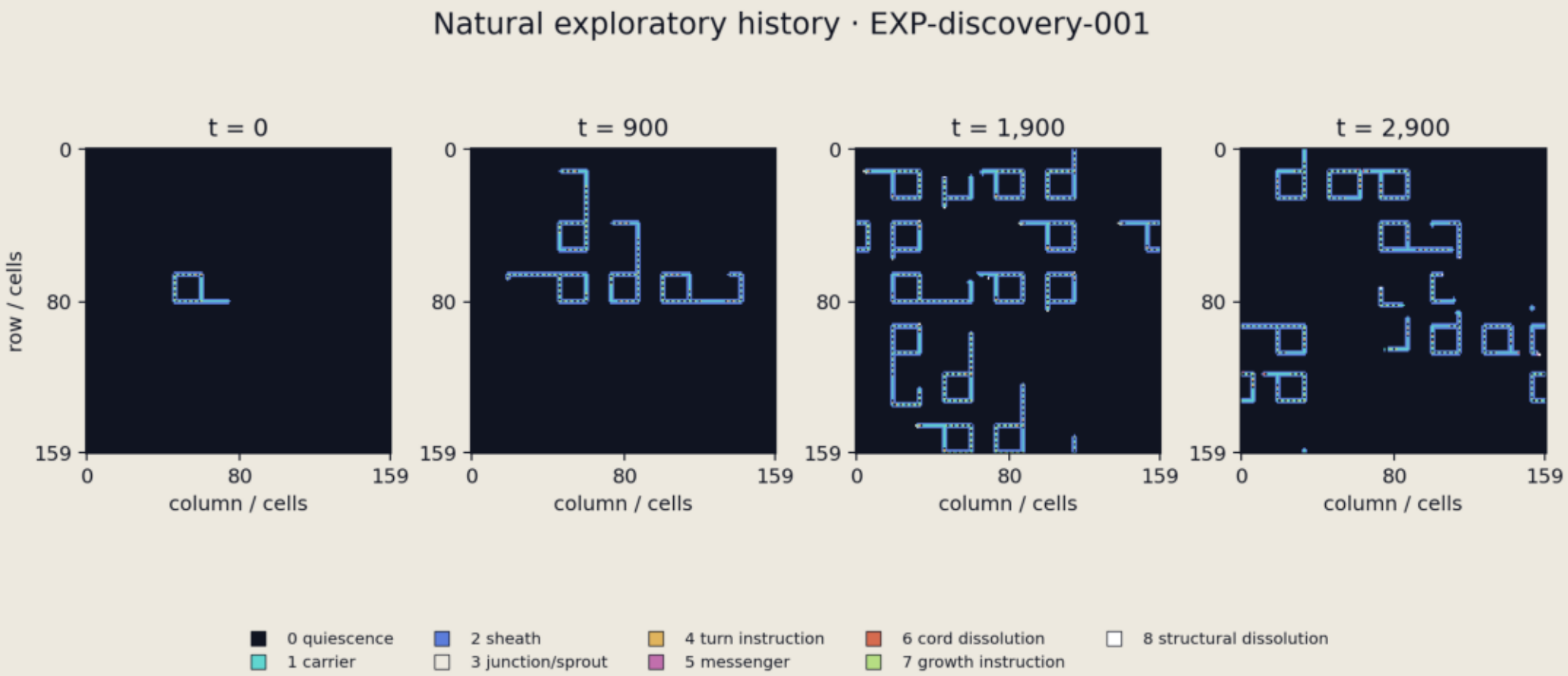



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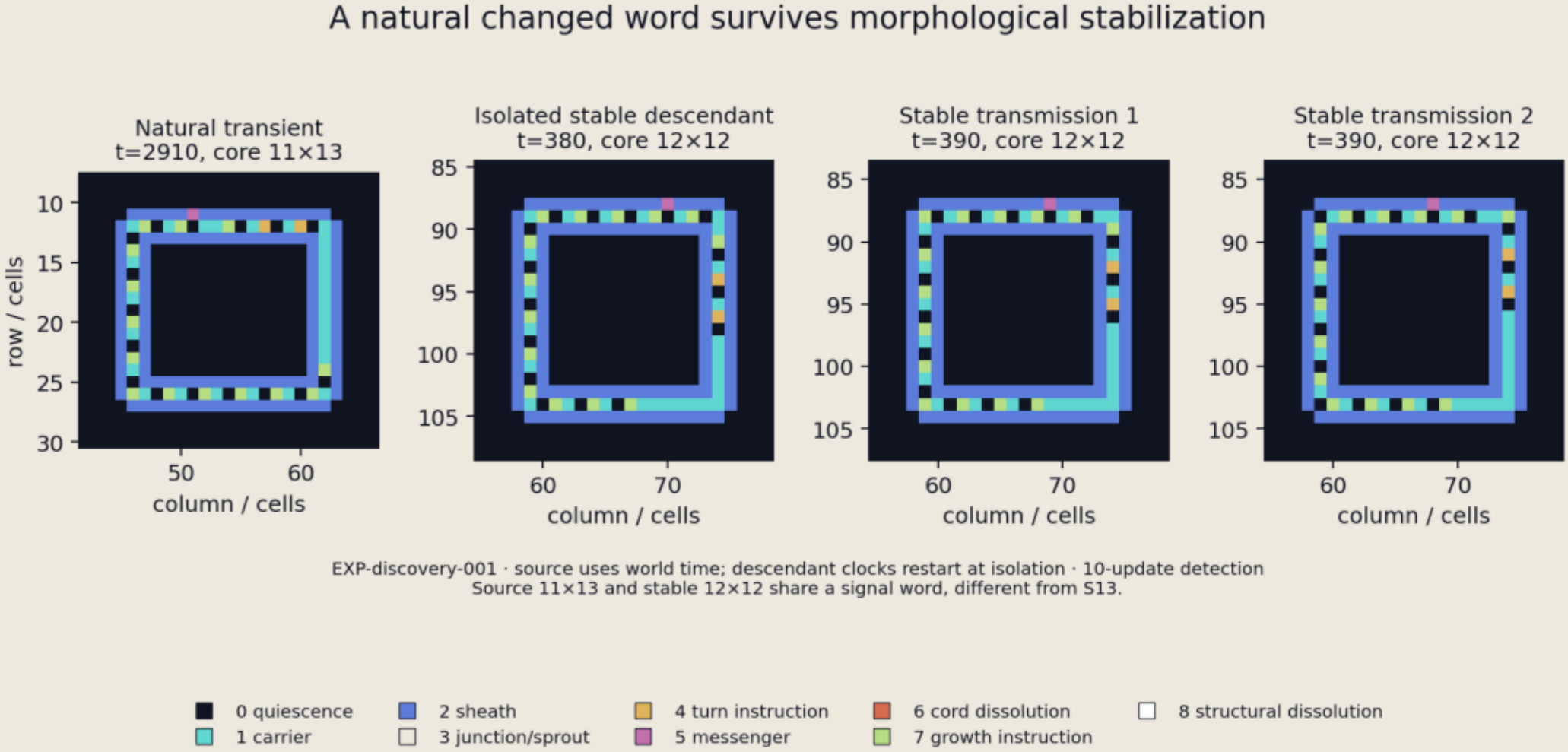



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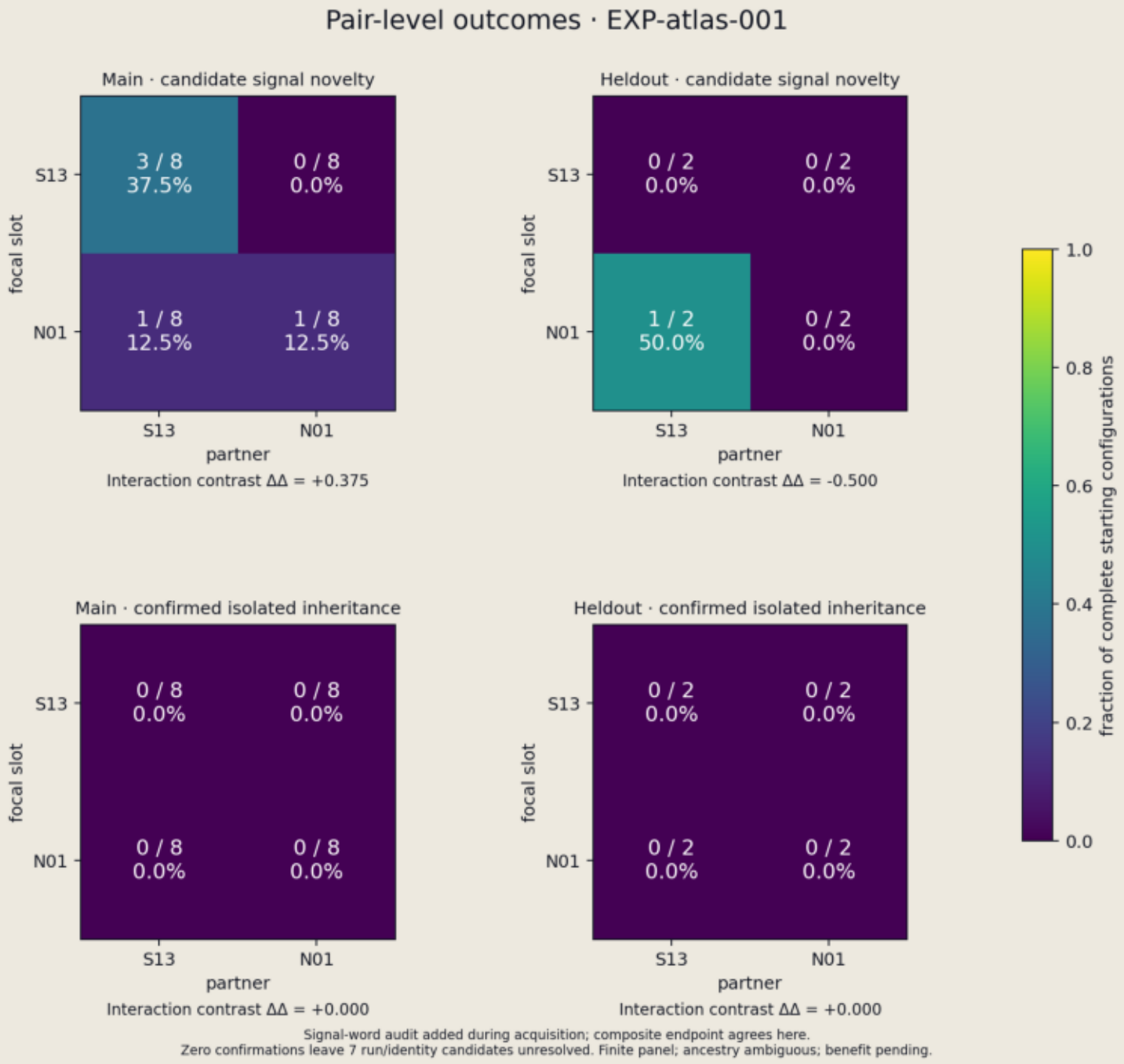



Figure: outcome-matrix. Source and regeneration: figures/manifest.json.

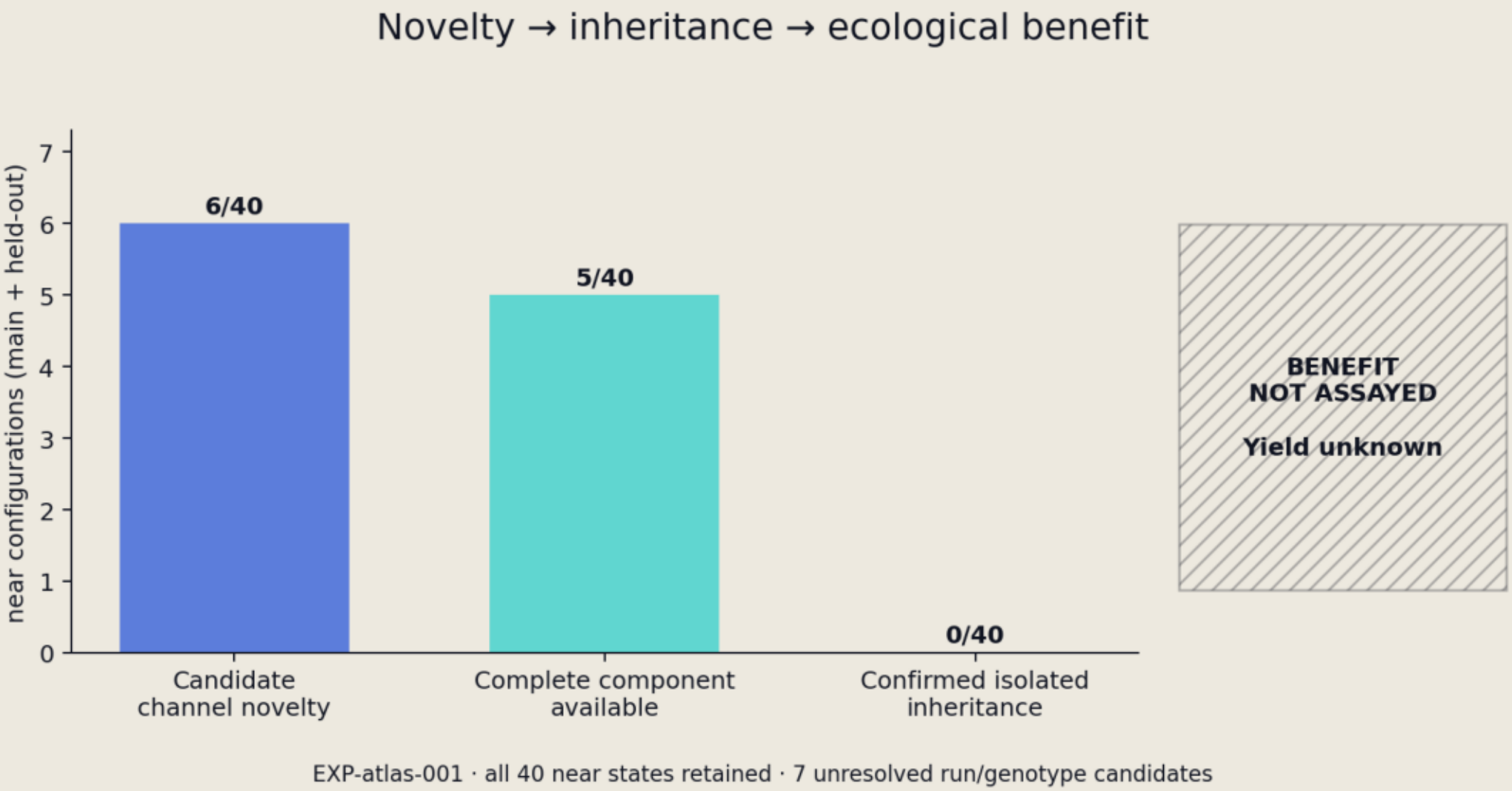



Figure: evidence-stages. Source and regeneration: figures/manifest.json.

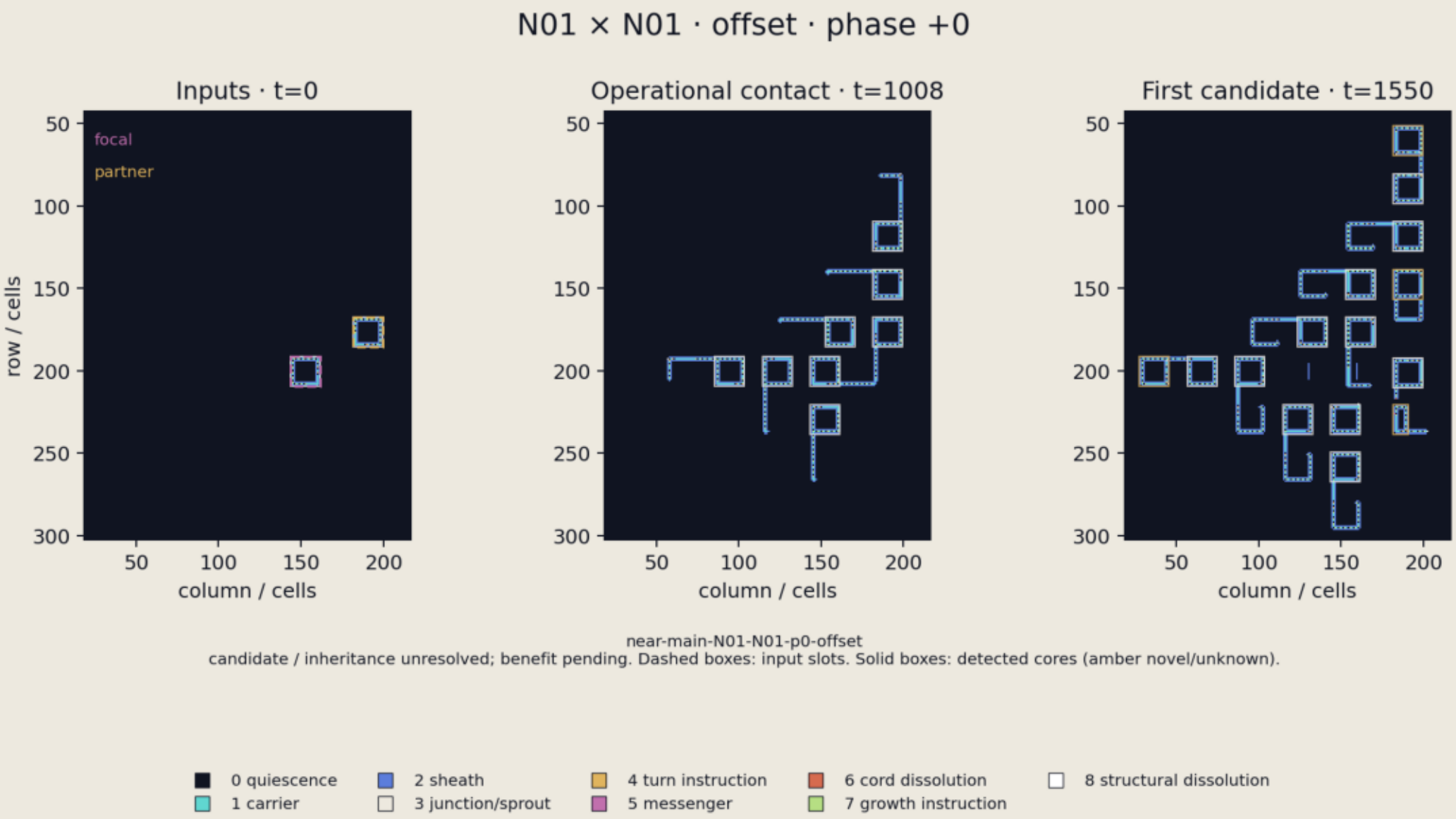



Figure: encounter-1. Source and regeneration: figures/manifest.json.

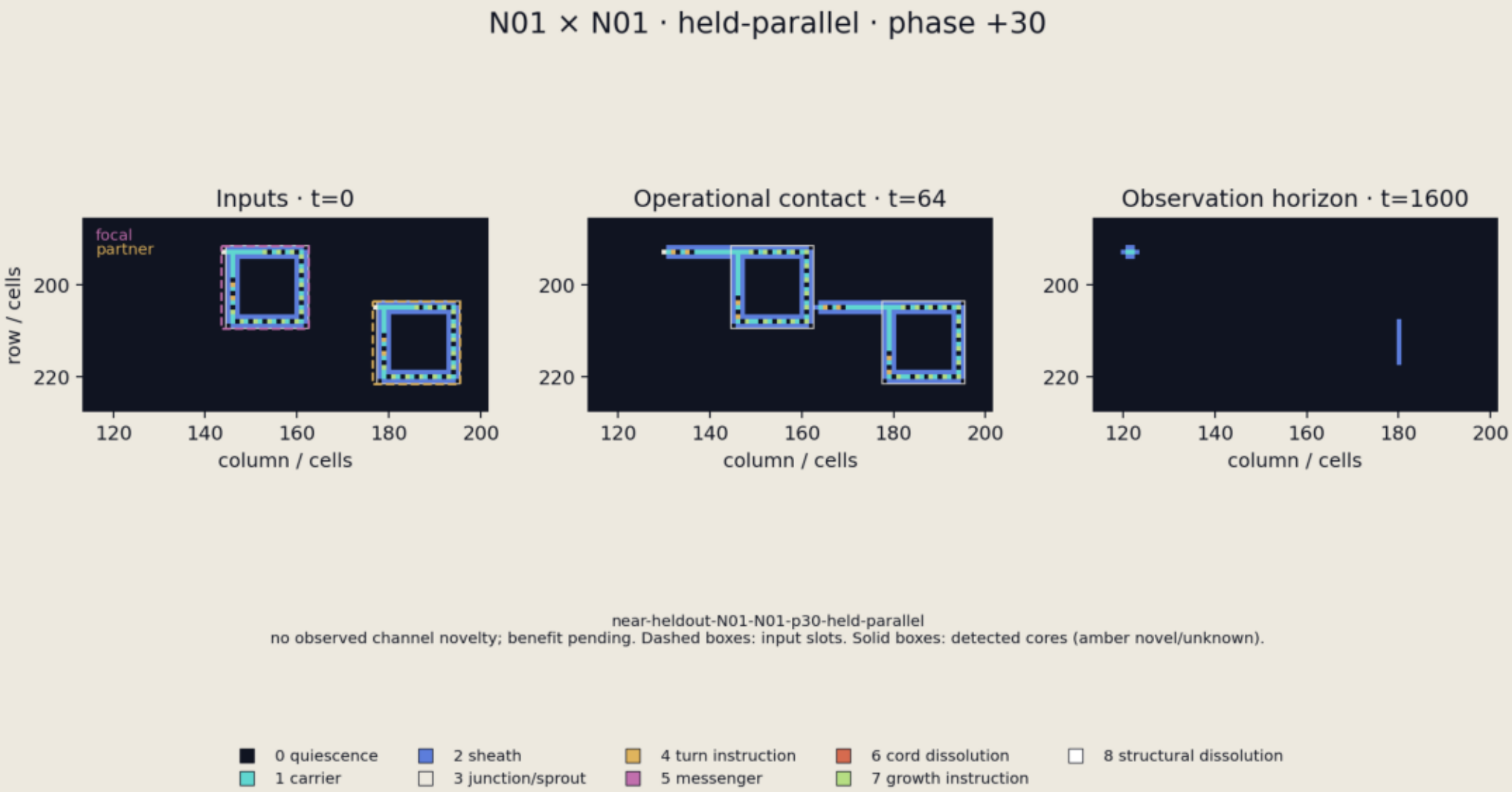



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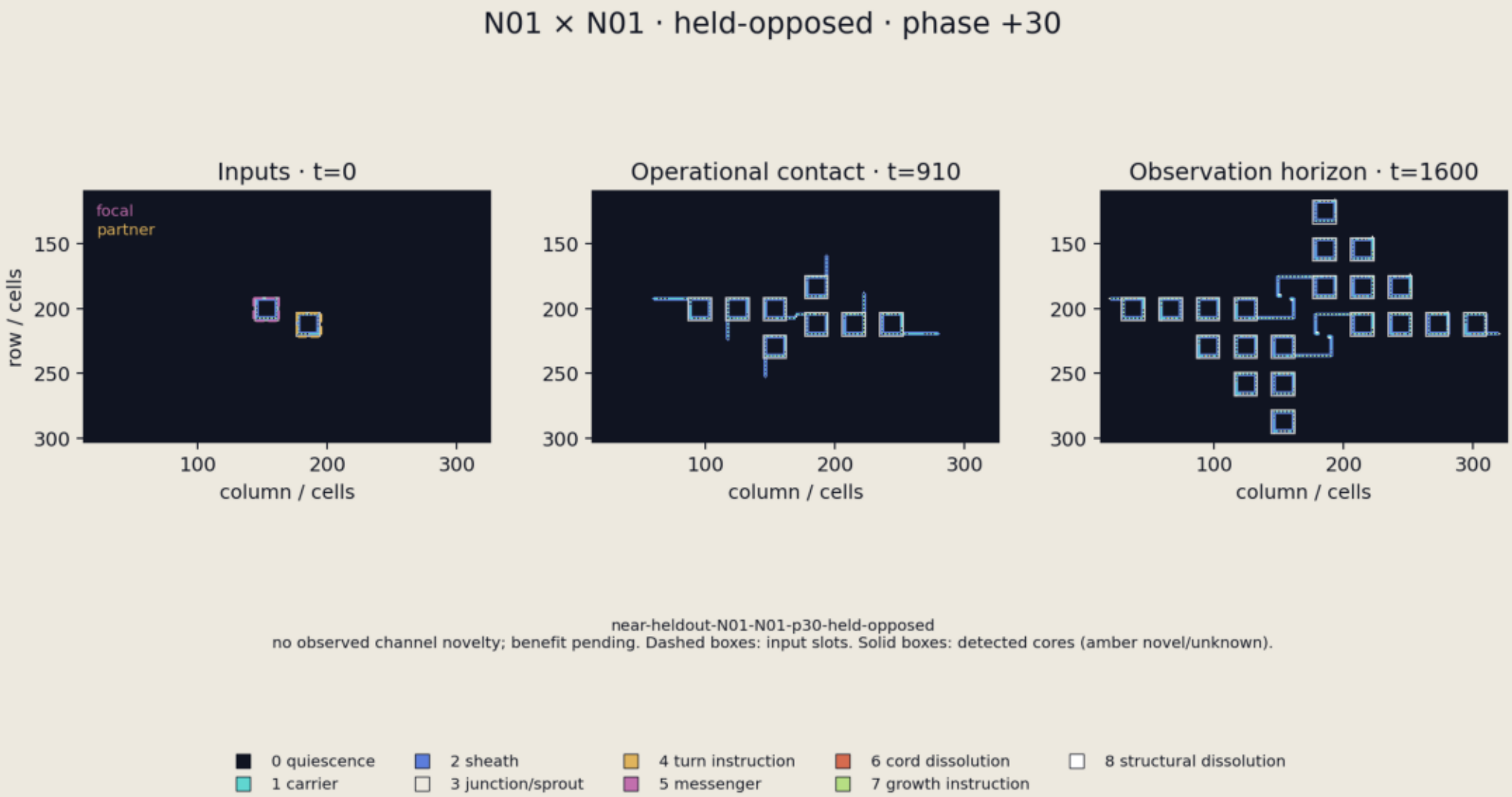



Figure: encounter-3. Source and regeneration: figures/manifest.json.